

Detached Attachment between ETFs and their underlying Indices: A Long Memory Perspective

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Abstract

This paper analyzes the long run relationship between domestic equity Index ETFs and their respective underlying Indices in various markets. In doing so, we study the inter-relationship between these ETFs and their underlying equity indices using the traditional Cointegrated Vector Autoregressive (CVAR) and Fractionally Cointegrated Vector Autoregressive (FCVAR) models. The FCVAR model is an extension of CVAR model that is geared towards capturing long memory in a multivariate framework. We examine if FCVAR is superior to the CVAR framework in terms of in-sample fit and out-sample forecasting performance. Our findings pinpoint the insignificant role played by Long Memory when it comes to modeling the relationship between domestic ETFs and their underlying indices. Lastly, our findings should nudge practitioners to exercise caution while engaging in arbitrage trades that involve domestic ETFs and their underlying indices.

Keywords: Long Memory, Exchange Traded Funds, Fractional Cointegration, Forecasting

JEL Classification: C58, G14, G15

Declarations of interest: none

1. Introduction

An Exchange traded fund (ETF) is a type of investment fund that replicates a market based index and is traded on stock exchanges ([Lettau & Madhavan, 2018](#)). Recent market statistics suggest that ETFs are an important class of investment instruments now. The assets under management (AUM) for open-ended funds worldwide was more than US\$54.8 trillion at the end of March 2020. ETFs accounted for more than 11% of this figure, amounting to an AUM of US\$6.16 trillion ([IIFA, 2020](#)). Moreover, about 98% of total ETF assets are index based instruments ([Sushko & Turner, 2018](#)).

This remarkable shift in investor preference towards passive index based investment instruments appears to be a new trend. Worldwide ETF assets are now about eight times the amount they were

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in 2008. US ETFs had a total AUM of more than US\$ 4.5 trillion at the end of July 2020 from about US\$ 2 trillion at the end of 2014 (ICI, 2020). In the larger segment of US equity funds, passive funds accounted for more than a 50% share of the AUM, with about US\$280 billion more than active funds at the end of April 2020, whereas in 2015, passive funds were behind active funds by more than US\$1.3 trillion (Thomas & Watson, 2020). Such trends are visible in other parts of the World as well. With the exception of Latin America, ETF AUMs in most other regions have grown at an average annual growth rate of about 20% since 2008 (ICI, 2019).

Studies on ETFs usually focus on the arbitrage mechanism between the ETF and its net asset value (NAV), wherein researchers measure the presence of any premium/discount in ETF market prices. ETFs have found to be better at following their underlying assets than other types of collective investment instruments, such as closed ended funds (Hughen & Mathew, 2009) and index mutual funds (Agapova, 2011). Within the ETF segment, it has been observed that domestic ETFs have smaller tracking errors than international ETFs (Engle & Sarkar, 2006; Tse & Martinez, 2007; Hilliard, 2014) due to confounding issues like exchange rate and different market dynamics.

Few studies seem to have studied the cointegration relationship between an ETF price and its underlying market index price. This is important in the light of an insightful work done by Dimpfl (2014) that suggests that for two indices to be cointegrated, they should be exactly same in terms of their constituents. He found that even US based indices like S&P500, NASDAQ and DJIA were not cointegrated. In the light of these findings, we believe that ETFs, that are designed to mimic their market equity indices would be a good candidate to study for such cointegration relationships (if any). Qadan & Yagil (2012) found the presence of cointegration between US based ETFs and their underlying indices based on the study period of 2006-2008. However, they also reported that tracking errors increased during times of financial uncertainties. In contrast, another study found that the top index equity ETFs in New Zealand are not well cointegrated with their underlying equity indices (Chen et al., 2017).

Our study aims to analyse prominent domestic equity index ETFs from various countries to measure for any cointegration relationship with their underlying indices. Additional criterion of full replication of underlying indices without any leverage or derivatives is imposed to prevent potential confounding effects. Subsequently, we also compare in-sample fit and out-of-sample forecasting performance with fractionally cointegrated vector auto-regressive (FCVAR) model which is an extension of the traditional cointegration (CVAR) framework. This is motivated by some related findings in literature that suggest the presence of non-linear dynamics in these instruments (Madhavan, 2014; Tiwari et al., 2017; Zhu & Bao, 2019). Similar studies on equity indices from various geographies have provided support for long memory in equity indices (Barkoulas & Baum, 1998; Hull & McGroarty, 2014). These findings motivate the question of whether long memory has any effect on the price dynamics of these instruments in a multivariate framework. This study has the following objectives:

1. To find whether ETFs are cointegrated enough with their underlying market indices.
2. If the FCVAR framework can provide more insights in terms of providing better in-sample fit using fractional cointegration (if any), and
3. To compare the out-of-sample forecasting performance of CVAR and FCVAR models.

2. Theoretical Framework

In the case of “traditional” cointegration (CVAR), each individual time series is assumed to be an $I(1)$ process, and the cointegrated vector (or the error correction term) is presumed to be $I(0)$. In other words, a time series vector, X_t , with components, say, $x_{1t} \sim I(1)$ and $x_{2t} \sim I(1)$, are said to be cointegrated of order $(1, 1)$ if there exists a vector $\beta = \begin{bmatrix} 1 & -\beta_1 \end{bmatrix}$, such that

$$z_t = \beta' X_t = [x_{1t} - \beta_1 x_{2t}] \sim I(0)$$

However, the Granger representation theorem that forms the basis for cointegration, only requires that the error correction term be stationary. The error correction term can react slower to shocks which means that any deviations from equilibrium will be more persistent. In other words, the cointegrating relationship can possess long memory. Hence, a shock would dissipate at a rate slower than the exponential decay of an ARMA process. It was in this context, that the idea of fractional cointegration was proposed (Granger, 1986). A time series vector, X_t , with components, say, $x_{1t} \sim I(d)$ and $x_{2t} \sim I(d)$, are said to be fractionally cointegrated of order (d, b) if there exists a vector $\beta = \begin{bmatrix} 1 & -\beta_1 \end{bmatrix}$, such that

$$z_t = \beta' X_t = [x_{1t} - \beta_1 x_{2t}] \sim I(d - b)$$

Johansen (2008) improved upon this idea by proposing the fractionally cointegrated VAR (FCVAR) model. The FCVAR model builds over the cointegrated VAR (CVAR) model of Johansen (1995) and more theory on this was developed in Johansen & Nielsen (2012). In this paper, we use the FCVAR framework proposed by Dolatabadi et al. (2016). The FCVAR specification used is as follows:

$$\Delta^d X_t = \alpha \Delta^{d-b} L_b (\beta' X_t + \rho') + \sum_{i=1}^k \Gamma_i \Delta^d L_b^i X_t + \xi + \epsilon_t,$$

where, $\epsilon_t \sim i.i.d.(0, \Omega)$ for p variables with Ω being positive definite, the fractional difference operator denoted by Δ^d , $L_b = 1 - \Delta^b$ is used to denote the fractional lag operator. ρ is interpreted as the mean of the long-run equilibrium and ξ signifies the deterministic trend. α and β are $p \times r$ matrices with $0 \leq r \leq p$ with r being the cointegration rank. They denote the long-run parameters as in CVAR model with $\beta' X_t$ as the long-run equilibrium relation. The parameters in α specify the speed of adjustment coefficients. The parameters Γ_i , $\forall i = \{1, \dots, k\}$ specify the short-run dynamics of the variables.

The economic relevance of studying fractional cointegration is that it implies that a long-run equilibrium relationship exists between the variables being studied. Fractionally integrated equilibrium errors,

while mean-reverting, show persistence during the short run. This flexibility in modeling mean-reversion behavior is important for a better analysis of long run relationship of the variables in question. In addition, we also compare the forecasting performance of the FCVAR and CVAR frameworks (Dolatabadi et al., 2018).

3. Analysis

3.1. Data Description

We selected the largest domestic equity ETFs (measured by AUM at end of 2019) ETFs from several developed and developing economies¹. Table 1 provides details of the chosen ETFs.

We take daily closing log prices of each ETF and its underlying equity index from July 2009 to January 2020². Figure 1 provides the time series plots of the log prices of the ETFs and their underlying indices.

Table 1: List of ETFs chosen

Region	Domestic ETF	Equity Index	Inception Date (ETF)
Americas			
Canada	iShares S&P/TSX 60 Index ETF	S&P/TSX 60 Index	Sep 28, 1999
United States	SPDR S&P 500 ETF Trust	S&P500 Index	Jan 22, 1993
Europe			
Eurozone	iShares Core Euro Stoxx50 ETF	Euro Stoxx 50 Index	Jan 03, 2001
United Kingdom	iShares Core FTSE 100 UCITS ETF	FTSE100 Index	Apr 27, 2000
Asia-Pacific			
Australia	Vanguard Australian Shares Index ETF	S&P ASX 300 Index	May 08, 2009
China	China 50 ETF	SSE 50 A Share Index	Feb 23, 2005
India	Nippon NIFTY BeES ETF	NIFTY50 Index	Jan 07, 2002
Japan	Nomura TOPIX ETF	TOPIX Index	Jul 13, 2001

A simple visual inspection seems to suggest the presence of trend in many cases. Preliminarily, we see the Indian ETF to diverge slightly from its index values at places. We expect to see lower level of cointegration in the Indian ETF.

3.2. Cointegration estimation

We test for cointegration relation (if any) between the domestic equity ETFs and their underlying indices using the CVAR model. Table 2 provides the results of this analysis.

We find weak evidence of cointegration between ETFs and indices for several cases. In the case of the Chinese ETF, it appears that there is no cointegration. There was weak evidence of cointegration for the Indian ETF. For the UK and USA ETFs, we see a full rank matrix which suggests that cointegration

¹For India, the largest ETF that has been available since July 2009 is chosen.

²Business cycle expansions and contractions as per NBER, USA (<https://www.nber.org/cycles/>)

Figure 1: Log Price plots of ETFs and their underlying indices from various markets

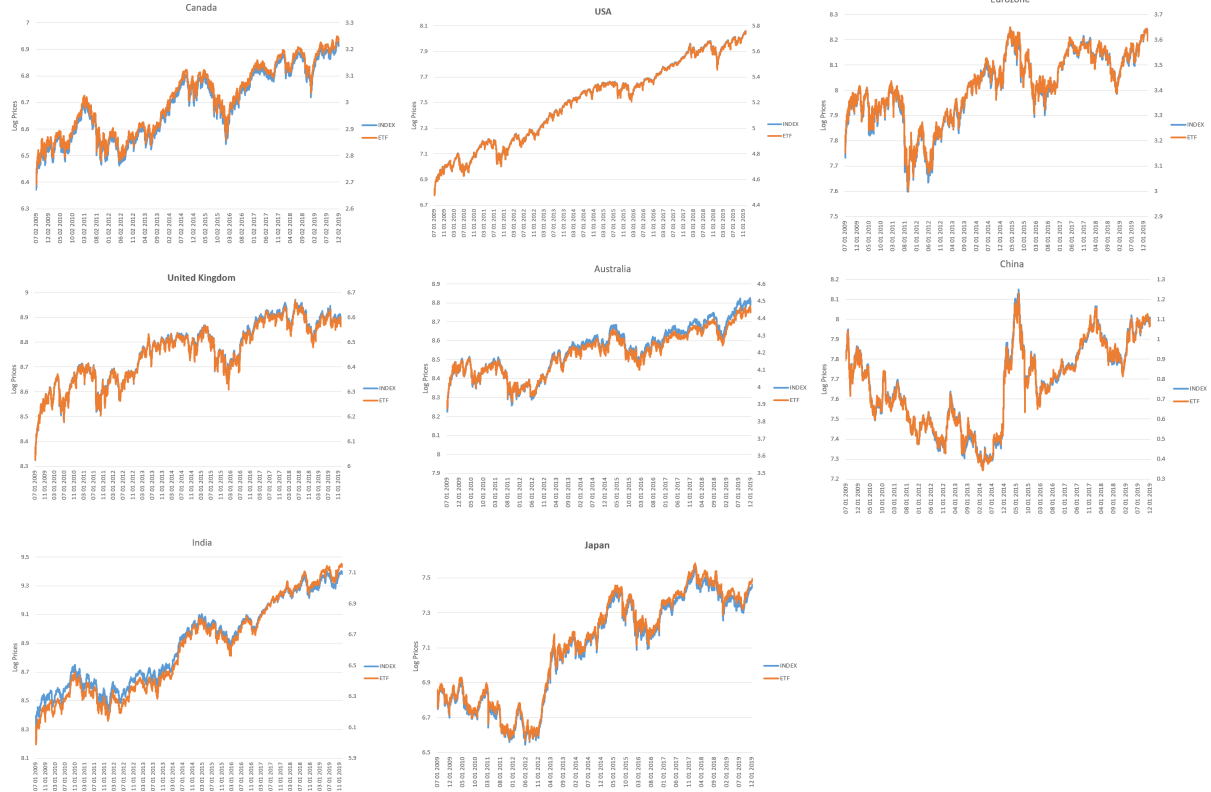


Table 2: CVAR Estimation

		Australia	Canada	China	Eurozone	India	Japan	UK	USA
ADF Test, $H_0 : I(1)$		can't reject	can't reject	can't reject	can't reject	can't reject	can't reject	reject	reject
KPSS Test, $H_0 : I(0)$		reject	reject	reject	reject	reject	reject	reject	reject
CVAR lag, k									
k (const)		8	4	8	6	9	6	5	30
CVAR rank Testing (Johansen's Method for Rank Estimation)									
ecdet (const)	$r = 0$	**	**		**	*	**	**	**
	$r = 1$							*	*
CVAR rank, r									
CVAR rank		1	1	0	1	1	1	2	1
Cointegrated Vector Estimates (for $r=1$)									
Constant, ρ	β_1 (Index)	1	1		1	1	1		1
	β_2 (ETF)	-1.026	-0.996		-1.032	-0.946	-0.980		-1.003
	ρ	-4.242	-3.693		-4.485	-4.814	-0.120		-2.286

^a Note to Table 2: Results of the ADF and KPSS tests are common for both the Index and the ETF. CVAR lag values (k) are selected using AIC.

* and ** denote rejection of the null hypothesis at 5% and 1% significance levels, respectively.

modeling is not required. This was found true for all the potential lags suggested at the lag selection stage. For the USA ETF, the evidence is mixed across different lag values. It appears that the confounding results for the UK and USA ETFs are due to the fact that these time series (both ETF and the Index) rejected the null hypothesis of the ADF test. This suggests the presence of long memory and we check if the FCVAR framework can provide more insights.

The trend component estimated as a part of the cointegrated vector was found to almost equal to zero. Hence, results are shown for the CVAR estimated with a constant term only. More results that form the basis for these choices are not reported for the sake of brevity.

3.3. Fractional Cointegration estimation

Log Prices are considered to follow random walk ($d = 1$). We cross-check this by estimating the fractional integration parameter (d) for log prices using the [Reisen et al. \(2017\)](#) methodology which was found to be robust to outliers in the data. Table 3 provides the results of this exercise. Most of the datasets support this choice of $d = 1$. In addition, based on the modeling assumptions used in the FCVAR framework, we choose to only estimate the value of the fractional cointegration parameter (b). Other parameters for FCVAR estimation were chosen as per [Dolatabadi et al. \(2016\)](#) and [Nielsen & Popiel \(2016\)](#).

Table 3: Estimates of fractional integration parameter (d), for log prices

Country/Region	ETF (log price)			Index (log price)		
	$\hat{d}$	se	$H_0 : d = 1$ (p -value)	$\hat{d}$	se	$H_0 : d = 1$ (p -value)
Australia	0.934	0.038	0.089	0.932	0.035	0.058
Canada	0.944	0.031	0.079	0.945	0.031	0.082
China	0.993	0.042	0.393	1.009	0.045	0.391
Eurozone	0.901	0.041	0.022	0.911	0.043	0.048
India	1.031	0.022	0.143	1.024	0.024	0.244
Japan	1.049	0.039	0.181	1.043	0.040	0.223
UK	0.966	0.037	0.262	0.966	0.037	0.262
USA	0.980	0.017	0.192	0.980	0.017	0.199

^a Note to Table 3: The bandwidth parameter is taken as 0.7 ([Reisen et al., 2017](#)).

Table 4 provides the FCVAR estimation results that include the estimated long memory of cointegrated vectors. The FCVAR output provides evidence in favour of the FCVAR model for all countries except Canada and Eurozone. For the Canadian and Eurozone ETF, the parameter b were estimated to be very close to 1 and hence the resulting model is very similar to the CVAR specification. For the UK

ETF, rank tests were rejected for both rank = 0 and 1, a case similar to what we found in the CVAR estimation. Hence, the FCVAR was not estimated for the UK ETF-Index pair. In addition, while the FCVAR estimates for the Indian ETF were found to be superior to the CVAR model, they failed one or other diagnostic tests (all lags) which also finds support in literature ([Madhavan & Maheswaran, 2016](#)). They are reported here to show the presence of long memory.

Table 4: FCVAR Estimation results

	Australia	Canada	China	Eurozone	India	Japan	USA
ADF Test – $H_0 : I(1)$	can't reject	can't reject	can't reject	can't reject	can't reject	can't reject	reject
KPSS Test – $H_0 : I(0)$	reject	reject	reject	reject	reject	reject	reject
CVAR lag, k	8	4	8	6	9	6	30
FCVAR lag, k	5	3	5	4	3	5	5
CVAR cointegration rank, r	1	1	0	1	0	1	1
FCVAR cointegration rank, r	1	1	1	1	1	1	1
FCVAR Estimation							
Level of Fractional integration, d	1	1	1	1	1	1	1
Level of Cointegration, b	1.245	0.906	1.556	0.953	0.885	1.221	1.348
se(b)	0.016	0.055	0.048	0.027	0.02	0.059	0.038
Long Memory of the equilibrium relationship, $I(d - b)$	-0.245	0.094	-0.556	0.047	0.115	-0.221	-0.348
H_0 : CVAR is better (p -value)	0.000	0.067	0.000	0.097	0.000	0.005	0.000
β_1 (Index)	1	1	1	1	1	1	1
β_2 (ETF)	-1.011	-0.998	-1.004	-1.033	-0.944	-0.983	-1.004
Constant (ρ)	-4.316	-3.692	-6.899	-4.513	-4.833	-0.094	-2.284
Robustness Checks							
Qu (2011) Test, H_0 : True Long Memory	can't reject	can't reject	can't reject	can't reject	can't reject	can't reject	can't reject
Wenger & Leschinski (2019) Test, H_0 : True Long Memory	can't reject	can't reject	can't reject	can't reject	can't reject	can't reject	can't reject
Wang et al. (2015) Test, H_0 : No Fractional Cointegration	reject	reject	reject	reject	reject	reject	reject
Souza et al. (2018) Test, H_0 : No Fractional Cointegration	reject	reject	reject	reject	reject	reject	reject

^a Note to Table 4: The Null Hypotheses for Robustness tests are checked at 5% level of significance.

3.4. Robustness checks

Subsequently, several tests are conducted for additional robustness checks. First, we test for the presence of spurious long memory owing to structural breaks using tests proposed by [Qu \(2011\)](#) and [Wenger & Leschinski \(2019\)](#). These tests check the Null Hypothesis of true long memory vs. the alternative of

spurious long memory due to structural breaks. We also conduct separate tests to check for the presence of fractional cointegration proposed by Wang et al. (2015) and Souza et al. (2018). The tests to check for spurious long memory showed no evidence of structural breaks for the ETFs and their Indices. In addition, the tests for fractional cointegration add support to the finding of fractional cointegration as per the FCVAR estimation.

3.5. Forecast Comparison

We compare forecast values for a horizon period of 1 month (daily, $h = 21$) using the FCVAR and the CVAR models. We compare this forecasting performance using two metrics. The first metric is the CW statistic that compares performance for nested models (Clark & West, 2007) with CVAR nested within the FCVAR model. The null hypothesis of this test statistic is that the CVAR forecasted values are at least as good as the ones from the FCVAR model, with the alternative being that the forecast values from FCVAR are better. In addition, we also report a modified Mean Absolute Scaled Error (MASE), originally proposed by Hyndman & Koehler (2006) defined as :

$$MASE = mean \left(\frac{\text{Absolute Percentage Error of FCVAR forecasts}}{\text{Absolute Percentage Error of CVAR forecasts}} \right)$$

MASE values < 1 indicate that the FCVAR forecasts values are better than CVAR model forecasts. Table 5 show the forecast comparison results. While literature suggests that the FCVAR framework can provide better forecasts (Dolatabadi et al., 2018), our findings suggest only marginal improvements (at best) from using the FCVAR model.

Table 5: Forecast comparison of FCVAR vs CVAR models

	CW Statistic		MASE	
	ETF	Index	ETF	Index
Australia	-1.494	-1.471	1.021	1.003
Canada	-0.214	0.069	0.997	0.996
China^{\$}	-1.207	-1.717	1.046	1.085
Eurozone	0.502	0.634	0.996	0.996
India	1.885*	1.915*	0.989	0.982
Japan	2.119*	2.131*	0.969	0.987
USA	0.196	0.294	1.011	1.011

Notes to Table 5: CW statistic > 0 favor FCVAR forecasts.

* denotes rejection of null hypothesis at 5% level of significance.

^{\$}: CVAR forecast values for China are calculated by forcing $r = 1$.

MASE values < 1 favor FCVAR forecasts.

4. Conclusion

The presence of long memory or anti-persistence depends on the range of integration order $I(d - b)$ of this ETF-Index cointegrated pair. For $0 < 1 - \hat{b} < 0.5$, there is long memory and anti-persistence for $-0.5 < 1 - \hat{b} < 0$. Put differently, the series will converge to equilibrium only very slowly under these two scenarios. Moreover, the fractional parameter b is significantly different from one for most cases, which suggests that the fractional cointegration model is more appropriate. Allowing fractional cointegration also requires fewer lags to be used in estimation. However, the forecasting performance of FCVAR model was not found to be statistically better than the CVAR forecasts.

The lack of cointegration between many domestic ETFs and their indices is surprising. Having said so, it needs to be noted that most ETFs (except ETF Trusts, such as the chosen US ETF) are allowed to use stock lending to improve returns. This “activeness” on the part of these ETFs can lead to increased risk for investors (Cheng et al., 2019). For example, the UK ETF (iShares Core FTSE100 ETF) has lent about 5% of its AUM at the end of June 2020³ and the proportion of stock lending has gone as high as 12.41% of its AUM historically. Further, the lack of cointegrating relationship between domestic ETFs and their underlying indices may be symptomatic of high market indexation (Baltussen et al., 2019), an ETF investor base skewed towards institutional investors and the investors’ inclination to use ETFs for short-term strategies (Broman & Shum, 2018; Ben-David et al., 2018), for these realities can impact volatility of underlying assets.

To sum up, this study provides the following takeaways. Domestic ETFs and their respective underlying indices possess long memory, as evidenced by the presence of fractional cointegration. Having said so, the application of FCVAR model does not bring about a significant improvement in forecasting performance of these cointegrating frameworks. These findings are pertinent to investors who consider ETFs as an avenue to hedge their market exposure and traders who execute pairs trading strategies that involve ETFs and their underlying indices. Such investors should exercise caution, for the data generating process (DGP) of an ETF may not always mimic the DGP of the underlying index.

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³[Homepage of the iShares Core FTSE 100 UCITS ETF](https://www.ishares.com/etfs/ftse100)

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